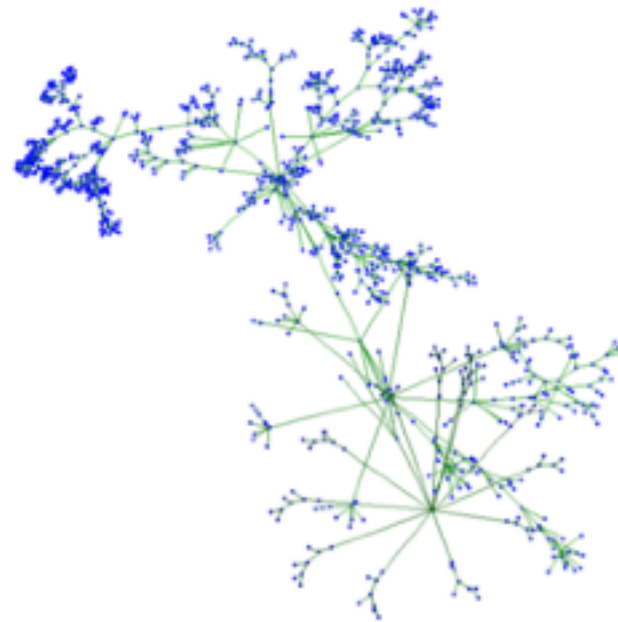


100% accurate, style-preserving transformation because of a tech leap in code representation

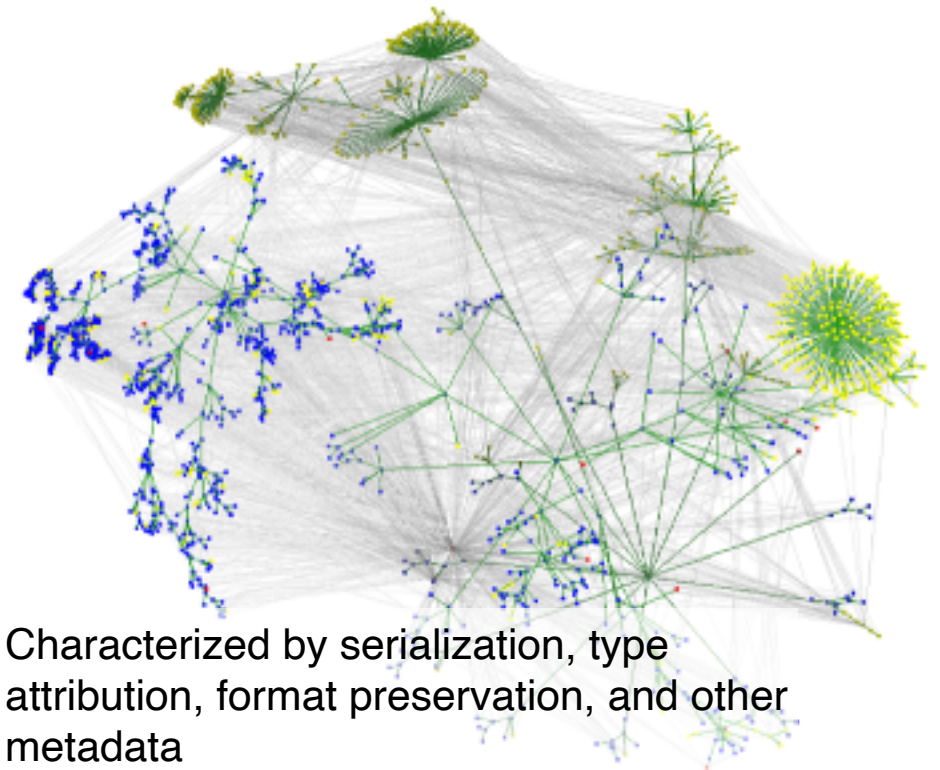
Text-based search

```
void m() {  
    if(c > 2) {  
        x = c;  
    }  
    while(c < 10) {  
        x += p();  
        c++;  
    }  
}
```

Abstract Syntax Tree (AST)



Lossless Semantic Tree (LST)



Characterized by serialization, type attribution, format preservation, and other metadata

+ another 600,000 nodes that didn't fit in this graphic

Imperative Recipe API

```
public class MakePublic extends Recipe {
    public String getDisplayName() {
        return "Make Class Public";
    }
    @Override
    protected JavaVisitor<ExecutionContext> getVisitor() {
        return new ChangeTypeVisitor();
    }
    private class MakePublicVisitor
        extends JavaVisitor<ExecutionContext> { /*...*/ }
}

class JavaVisitor<P> extends TreeVisitor<J, P> {
    J visitStatement(Statement statement) {}
    J visitAnnotatedType(J.AnnotatedType annotatedType) {}
    J visitAnnotation(J.Annotation annotation) {}
    J visitAssert(J.Assert azzert) {}
    J visitAssignment(J.Assignment assign) {}
    J visitAssignmentOperation(J.AssignmentOperation assignOp) {}
    //...
}
```

